Introduction

Introduction to Data Mining

Seoul National University

Spring 2026

Outline

1. **<u>Introduction</u>**
2. Course logistics
3. Basic concepts

Data is Power

Source: Popular Science

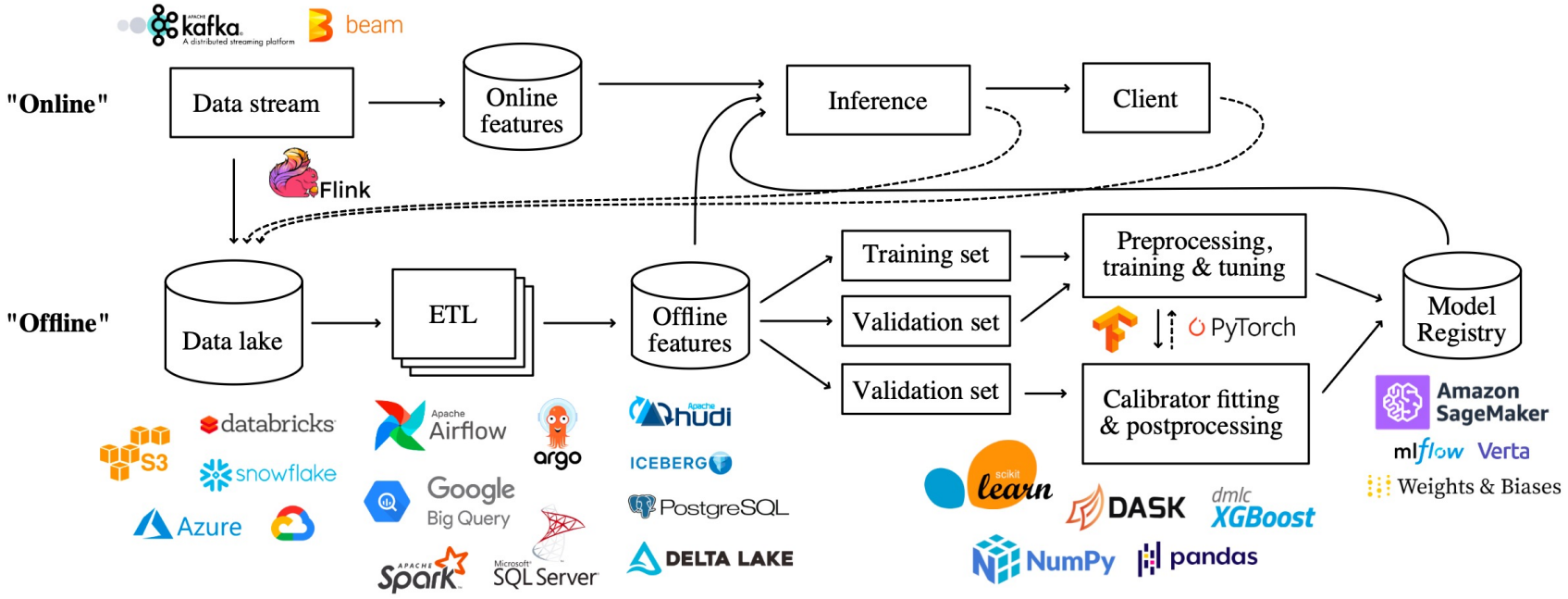
Data Mining

- Data creates value only when it can be turned into decisions.
- **Data mining** is a process of transforming **raw data** to **insights.**
  1. Storing data (in memory, on disk, or in the cloud)
  2. Managing data (using proper data structures)
  3. Analyzing data (using proper algorithms for specific purposes)

$$\text{Data Mining} \approx \text{Data Science} \approx \text{Data-Centric AI}$$

Data Mining Pipeline

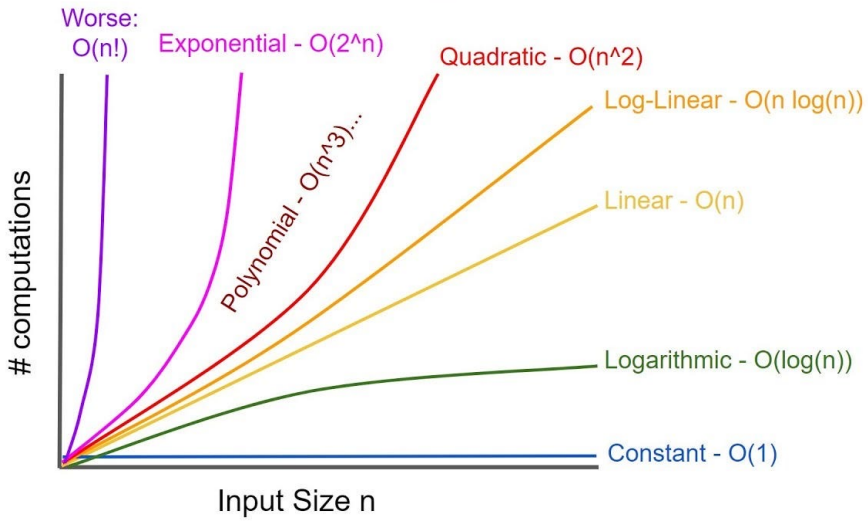
- Industrial data-processing pipelines are incredibly complex.
  - Real-world data is large, heterogeneous, and often streaming.



Source: Shankar and Parameswaran (2021)

What This Course is About

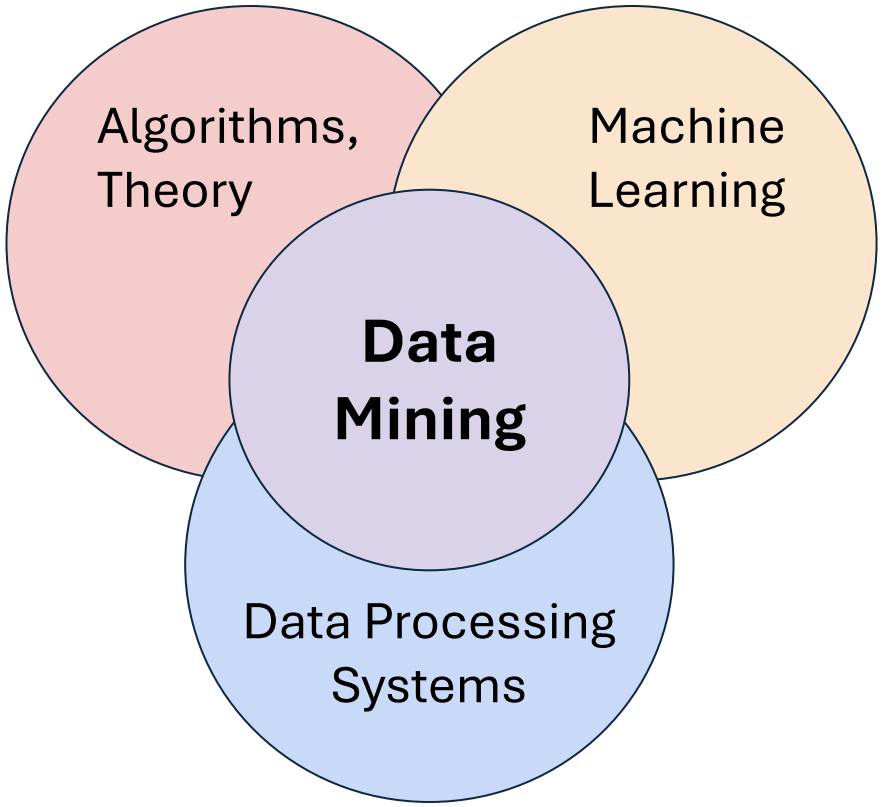
- **Goal:** Extraction of actionable information from large datasets.
- It's not all about **machine learning,** but most of it is!
- We emphasize algorithms that **scale,** often with parallelization.



Source: Medium

What We Cover

- Combines the best of algorithms, machine learning, and systems.
- No specific prerequisites, but
- Will challenge **a variety of skills:**
  - Mathematical intuition.
  - Problem-solving ability.
  - Implementation skills.



What Will We Learn?

- We will learn to **analyze different types of data:**
  - High-dimensional tabular data
  - Graph-structured data
  - Streaming (never-ending) data
- We will learn to **use different models of computation:**
  - Distributed computing
  - Single machine in-memory
  - Streams and online algorithms

What Will We Learn?

- We will learn to **solve real-world problems:**
  - Recommender systems
  - Market basket analysis
  - Filtering and clustering
- We will learn **various tools and techniques:**
  - Linear algebra (SVD, recommender systems)
  - Optimization (stochastic gradient descent)
  - Dynamic programming (frequent itemsets)
  - Hashing (LSH, bloom filters)

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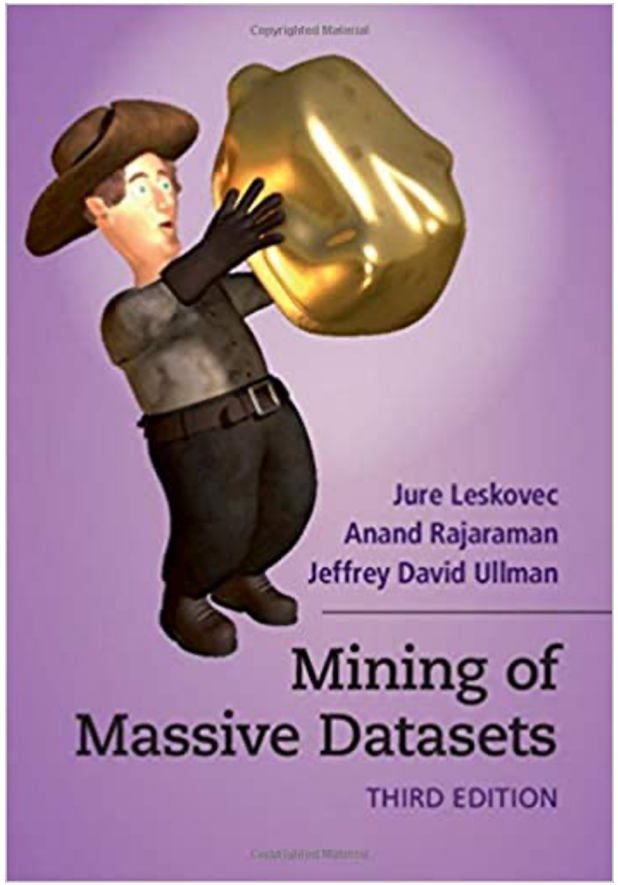
Staff

- **Lectures:** Tue/Thu 9:30 – 11:00
- **Location:** 302-107
- **Instructor:** Jaemin Yoo (301-406)
- **Teaching assistants:**
  - Jinmo Lee, Doyun Choi, Hak-Beom Kim, Minho Jeong
- **Contact:** snucse.idm@gmail.com

Course Materials

- **Textbook:** Mining of Massive Datasets
  - Available online: http://mmds.org.
  - Some lecture topics are not in the textbook.
- **Lecture slides:**
  - Will be uploaded to ETL prior to class.
- **Lecture videos (TBD):**
  - Will be recorded and upload to ETL after class.



Communication

- **ETL:**
  - For lecture slides, videos, and assignments.
- **Classum (TBD):**
  - For all questions and public communication.
  - For official course announcements.
- **Emails:**
  - Always use snucse.idm@gmail.com.
  - Only ask questions not suitable for Classum.

Course Work and Grading

- Assignments (40%)
- Midterm (30%)
- Final exam (30%)
- Participation (up to extra 2%)

Assignments

- 4 assignments on Python and Apache Spark programming.
- Due at 11:59pm.
- HW0 will be posted next week.
- HW0 is mostly a tutorial for Apache Spark (no evaluation).
- Each assignment will have specific instructions of what to do.

Tentative Class Schedule

Homework	Date Out	Date In
HW0	Week 2	-
HW1	Week 3	Week 6
HW2	Week 6	Week 9
Midterm	Week 7 or 8	
HW3	Week 9	Week 12
HW4	Week 12	Week 15
Final	Week 15	

Late Day Policies for Assignments

- **Assignments 0 – 3:**
  - 3 late days (in 24-hour chunks) can be used for any of these assignments.
  - Once the late days are exhausted, a score is reduced by 20% per late day.
  - No assignment will be accepted more than 4 days after its due date.
- **Assignment 4:**
  - No late submission is allowed as we need to submit grades to the system.

Exams

- **Midterm:**
  - Covers the first half of course
  - Other details (including the date) TBD
- **Final exam:**
  - Covers the second half of course
  - Other details (including the date) TBD

Tentative Course Outline

- **Midterm:**
  - MapReduce and Spark
  - Frequent itemsets
  - Finding similar items
  - Clustering
  - Dimensionality reduction
  - Recommender systems
- **Final:**
  - Link analysis
  - Neural nets and deep learning
  - Graph representation learning
  - Graph neural networks
  - Mining data streams

Class Participation

- We will not check attendance.
- Instead, we encourage you to answer questions on Classum.
  - We will give credit to both the # of answers and # of likes you receive.
- You also can get credit by reporting bugs in course materials.

Prerequisites

- Programming (in Python)
- Data structures and algorithms
- Linear algebra
- Basic probability theory
- Familiarity with writing proofs

What If I Don’t Know All This Stuff?

- Each topic is important for a small part of the course.
  - If you are missing an item, you can learn the material as you go:
    - **Linear algebra:** https://web.stanford.edu/class/cs246/handouts/CS246_LinAlg_review.pdf
    - **Probability:** https://web.stanford.edu/class/cs246/handouts/CS246_Proof_Probability.pdf
- The exception is programming.
  - To do well in this course, you need to be comfortable with writing code.

Course Policy

- Please use AI to support your learning and understanding.
  - AI can be a better teacher than almost any expert.
- **However, using AI for programming is not recommended.**
  - You will eventually use AI as a programmer.
  - But if you don’t know how to code, you can’t debug AI’s code.
  - This course is where you practice and develop your own skills.

Course Policy

- Nevertheless, you are free to use AI for any purpose.
  - Please explain how you used AI and what help it provided.
  - No deduction in points **as long as you are transparent.**
- You may also discuss assignments with others.
  - As long as you include their name(s) on the work you submit.

Workload

- This course will be **fast-paced** to many students.
  - We will cover a lot of topics for various problems.
- **Course time commitment:**
  - You may need to review the content after each lecture.
  - Each assignment may take >20 hours towards the end.
- You may consider dropping this course if not ready.

Programming Environment

- Your own computer.
- Lab machines in the department (TBA).

Acknowledgements

- Some slides of this course are adapted from or inspired by:
  - [CS246] Mining Massive Datasets from Stanford CS.
  - [EE412] Foundation of Big Data Analytics from KAIST EE.

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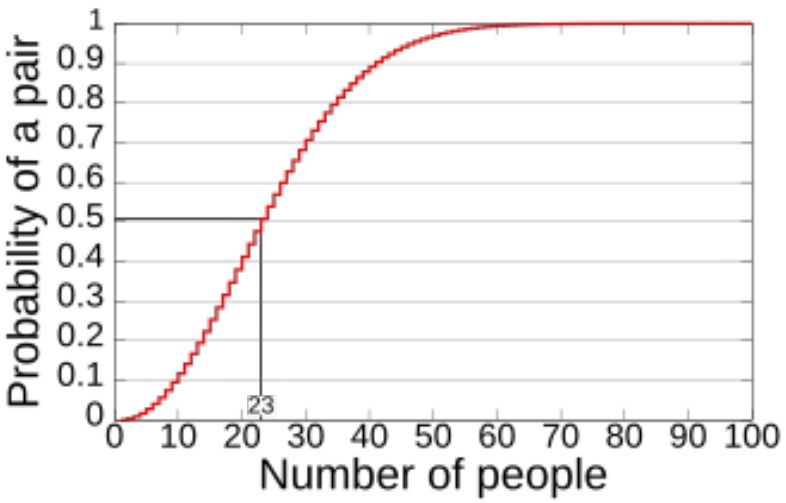
Statistical Limits on Data Mining

- An analyst can always discover patterns that are meaningless.
- **Bonferroni's principle:**
  - You look for events of a certain type within a certain amount of data.
  - Assume random data of the same size.
  - Calculate the expected number N of occurrences in the random data.
  - If N is large, then you must expect almost anything to be bogus.

Example: Birthday Problem



- There are 23 people in the room.
- Surprisingly, two share the same birthday!
- Seems like a surprising pattern … but is it?
- In fact, not really, since the probability is quite high (> 0.5).
- Almost half of all such groups will show the same pattern.

When Machine Learning is Useful

- Not all data mining algorithms use machine learning (ML).
- ML is useful when we are not sure what we are looking for.
  - Examples: recommendation, forecasting, and embedding.
- ML is not necessary when we can describe the goal directly.
  - Examples: locating people's resumes on the web.

Importance of Words in Documents

- **Problem:** Categorizing documents by their topic.
  - Document is a sequence of words.
- **Assumption:** Topics are identified by special words:
  - Baseball: “ball,” “bat,” “pitch,” “run,” and so on.
- **Observations:**
  - Frequency ≠ significance (e.g., “the” or “and”).
  - Rarity ≠ significance (e.g., “notwithstanding” or “albeit”).

Importance of Words in Documents

- **Idea:** Concentration of useful words in just a few documents.
- **TF-IDF** (Term Frequency times Inverse Document Frequency).
  - We are given a term i in a document j.
  - $\mathrm{TF_IDF}(i, j) = \mathrm{TF}(i, j) \times \mathrm{IDF}(i)$.
  - $\mathrm{TF}(i, j) \propto$ The number of occurrences of a term i in a document j.
  - $\mathrm{IDF}(i) \propto 1$ / The number of documents where a term i appears.

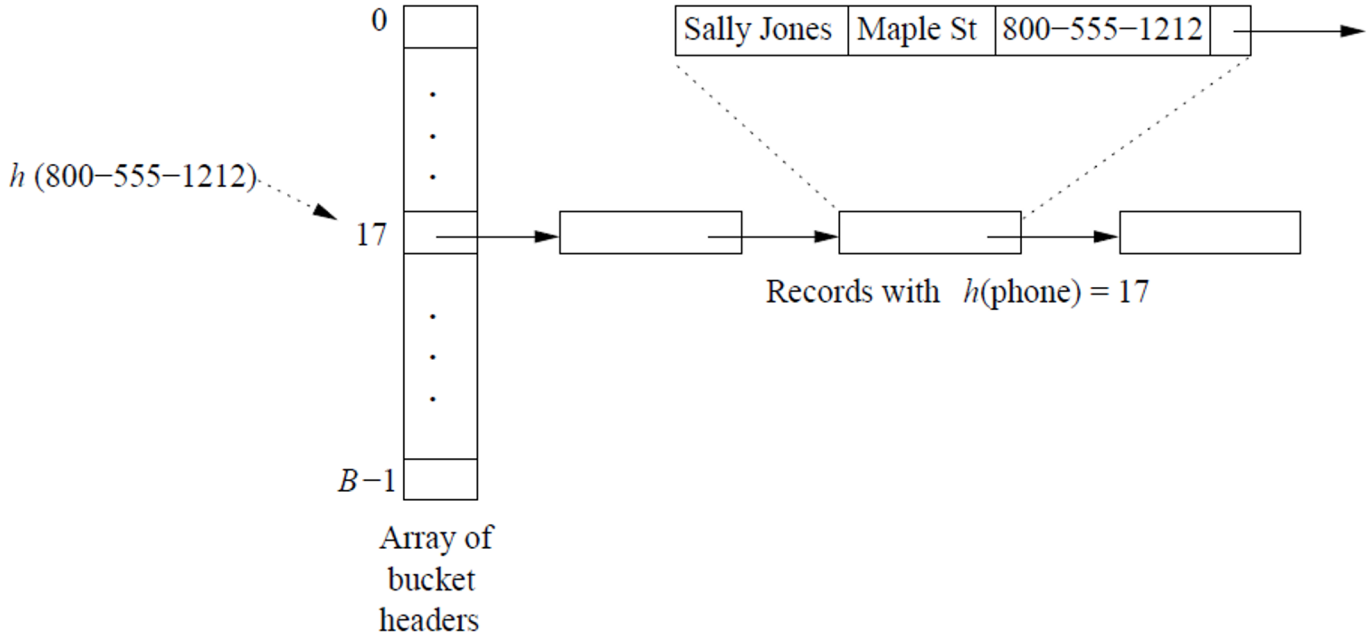
Hash Functions

- **Hash functions** are one of the most important concepts in CS.
- A hash function h takes a **key** and produces a **bucket number.**
  - The bucket number is an integer, say from 0 to $B - 1$.
  - Normally, a hash key is an integer, and B is a prime number.
- Enables us to build a constant-time data structure for big data.

$$\text{key} \longrightarrow \text{h(key)}$$

Indices

- An **index** is a data structure for retrieving records efficiently.
- E.g., a hash table where phone numbers are hashed to buckets.

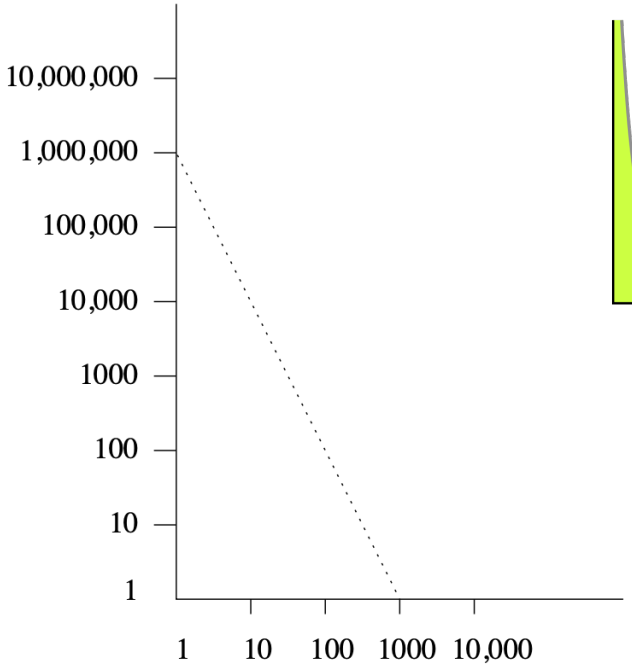


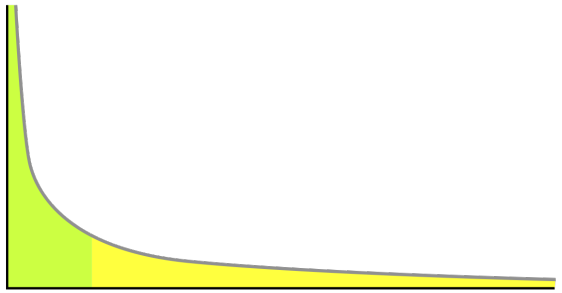
Hash Functions: Questions

- **Q1:** What if hash keys are not integers?
  - All data types have values that are composed of bits.
  - For strings, there are many ways to convert them to integers.
  - For complex types, recursively convert values to integers.
- **Q2:** Can we avoid collisions at all?
  - No, there always can be collisions.
  - There are many techniques to reduce collisions though.

Power Laws

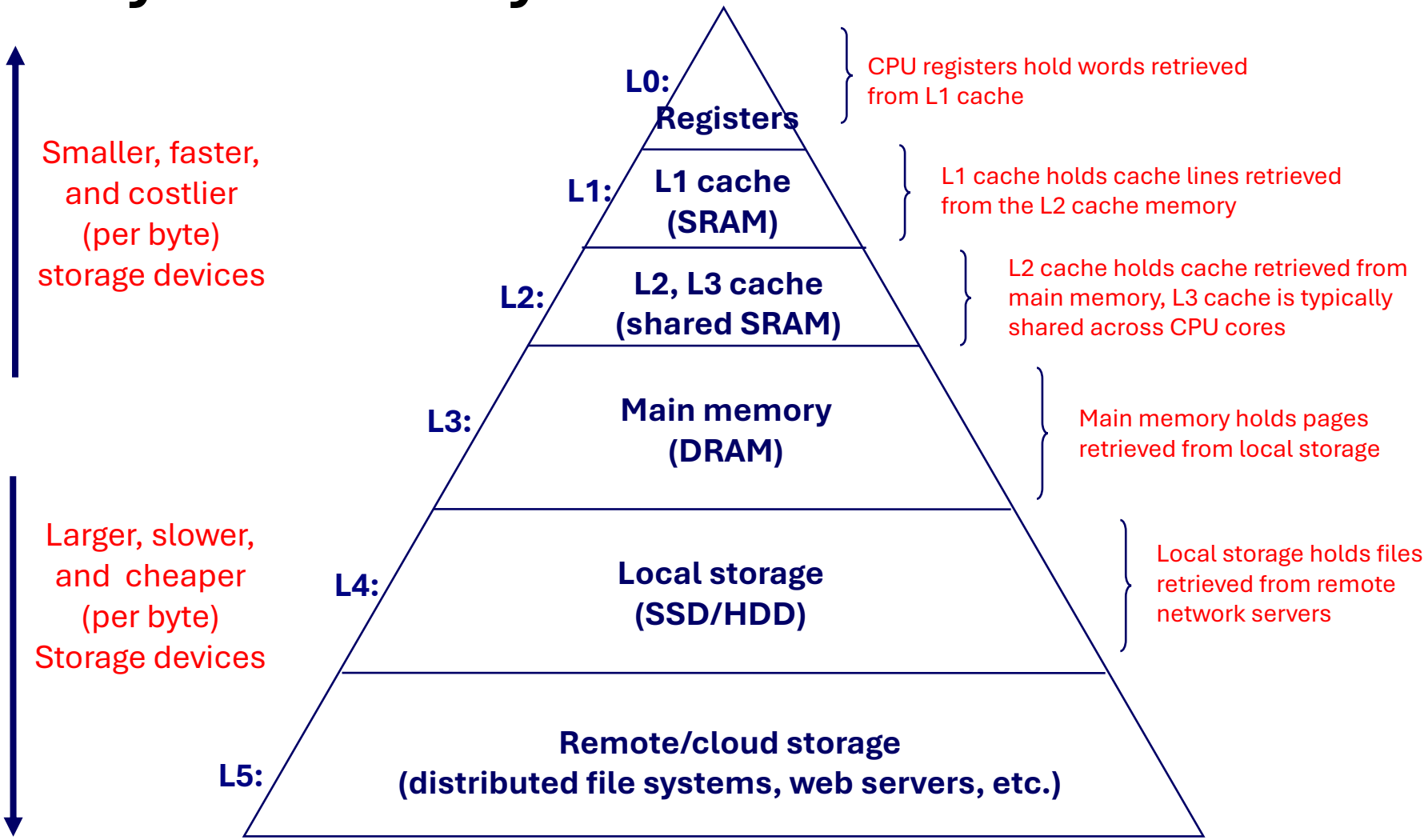
- Many two variables are related by a **power law.**
  - That is, a linear relationship between the logarithms of the variables.
  - General form: $\log y = a \log x + b$
- **Examples:**
  - Sales of products.
  - Sizes of websites.
  - Wealth of people.





In the linear scale

Memory Hierarchy



Summary

1. Introduction
2. Course logistics
   - Course materials
   - Communication
   - Course work and grading
   - Prerequisites
3. Basic concepts
   - Statistical limits
   - Basic knowledge of CS